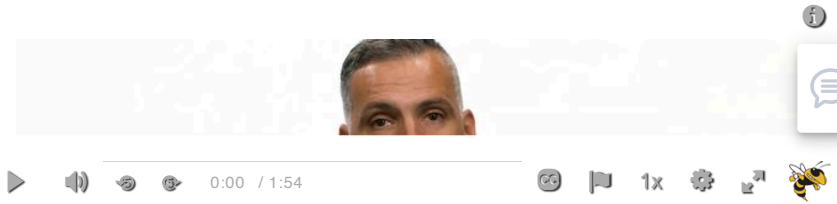


Lesson Nine Overview



Throughout the lecture, we are giving you some **“food for thought”** -- these are questions we suggest you think about as you go through this lesson. You do not need to submit solutions.

Learning Objectives

Students will be able to:

- Define what **“network contagion”** means
- Analyze standard network epidemic models in the context of realistic network models
- Explain how the structure of a network can have a major effect on the function and dynamic activity on that network
- Understand the role of superspreaders in epidemics

Required and Recommended Reading

Required Reading

- Chapter 10 -A-L. Barabási, [Network Science \(http://www.networksciencebook.com/\)](http://www.networksciencebook.com/), 2015 (available [online](http://www.networksciencebook.com/) [\(http://www.networksciencebook.com/\)](http://www.networksciencebook.com/))

Recommended Reading

- [Super-spreaders in infectious diseases](https://doi.org/10.1016/j.ijid.2010.06.020) [\(https://doi.org/10.1016/j.ijid.2010.06.020\)](https://doi.org/10.1016/j.ijid.2010.06.020) [Richard A.Stein \(Links to an external site.\)](https://www.sciencedirect.com/science/article/pii/S1201971211000245) [\(https://www.sciencedirect.com/science/article/pii/S1201971211000245\)](https://www.sciencedirect.com/science/article/pii/S1201971211000245), *International Journal of Infectious Diseases*, August 2011
- [Cellular Superspreaders: An Epidemiological Perspective on HIV Infection inside the Body](https://doi.org/10.1371/journal.ppat.1004092) [\(https://doi.org/10.1371/journal.ppat.1004092\)](https://doi.org/10.1371/journal.ppat.1004092) Kristina Talbert-Slagle et al., 2014

Assignments

L9: Quiz

L9: Knowledge Check